

The Myhill-Nerode Theorem

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1.1 Pumping

The pumping lemma is not a perfect characterization of non-regular languages. There exist languages which can satisfy the conditions of the pumping lemma, but be non-regular.

1.2 Myhill-Nerode

Lucky for us, there does exist a perfect characterization. It helps us prove non-regular languages to be non-regular, or regular languages to be regular. It also implies a unique and minimal¹ DFA for each regular language.

To give a high level idea, for any DFA D , and two strings which end on the same state in D , say x, y . Since it is deterministic, there is one outgoing transition from that state on 0, so $x0, y0$ will also end in the same state given x, y ended in the same state. We can abuse this! Lets generalize.

For a language L , let $\sim_L$ be an equivalence relation on Σ^* defined by $x \sim_L y$ if for all $z \in \Sigma^*$, $xz \in L \iff yz \in L$. Note that taking $z = \varepsilon$ shows that, if $x \sim_L y$, then either both x and y are members of L , or neither is.

Theorem 1.1 L is regular if and only if $\sim_L$ partitions Σ^* into a finite number of equivalence classes.

- ($\implies$) Suppose L is regular, and thus has a DFA to decide it, lets denote as D . For each $x \in \Sigma^*$, running x on D will stop in some state. Let $[q_i]$ be the set of all strings which stop on state q_i . Notice there are a finite number of such sets, one for each state, and $\Sigma^* = \cup_{i=1}^n [q_i]$, with each disjoint². Let $x, y \in [q_i]$, and let $z \in \Sigma^*$, and let q_j be the state D finishes if it starts in q_i and reads z . Then xz and yz both cause the machine to end in q_j . If q_j is an accept state, $xz, yz \in L$; if q_j is a non-accept state, $xz, yz \notin L$. This implies $xz \in L \iff yz \in L$ whenever $x \sim y$. Each set $[q_i]$ is then an equivalence class of $\sim_L$, and there are a finite number of them.
- ($\impliedby$) Let $\sim_L$ partition Σ^* into a finite number of equivalence classes. We can construct a DFA to decide L . Recall a DFA has the form $(Q, \Sigma, \delta, q_0, F)$. Let $q_i \in Q$ be the states and $[q_i]$ be the equivalence class for q_i .
- Q : Form n states, one for each equivalence class of $\sim_L$.
 - Σ : is the alphabet of L .
 - δ : For each $q_i \in Q$, $a \in \Sigma$, and $x \in [q_i]$, define $\delta(q_i, a) =$ the state for the class which contains xa .³

¹with respect to the number of states

²And for q_f the final state, $L = [q_f]$

³For this function to be well defined, we need that $x \sim_L y \implies xz \sim_L yz$. Do you see why?

- q_0 : Let the start state be the state for the equivalence class which contains ε
- F : For each state $q_j \in Q$, determine if an element of L is contained in the equivalence class for q_j .

This is a well defined DFA, which decides L . This implies that L is regular.

1.3 Usage

To use the theorem to prove a language is regular, show that $\sim_L$ exhibits a finite number of equivalence classes.

As an example, we prove $\{1^{2n} \mid n \in \mathbb{N}\}$ is regular⁴. Consider the sets $[q_0], [q_1], [q_2]$ where $[q_2]$ is all string containing at least one zero, $[q_0]$ is all even strings of ones, and $[q_1]$ is odd strings of ones. Notice that each of these sets partition $\Sigma^* = [q_0] \cup [q_1] \cup [q_2]$, and they are pairwise disjoint. For any $x, y \in [q_2]$ then $xz, yz \in [q_2]$ since they still contain a zero. For x, y both even or both odd length, and z no zeroes, then the parity of xz, yz is the same if the parity of x, y is. If z has a zero, then xz, yz again both are in $[q_2]$.

To use the theorem to prove a language is non-regular, give an infinite set S of strings, such that for each pair of strings $x, y \in S$ there is a least one string z such that $xz \in L, yz \notin L$ or vice versa. Then each string of S must belong to a separate equivalence class of $\sim_L$, so if S is infinite, there are infinitely many equivalence classes. Note that you may have a different z for each pair x, y .

As an example, we prove $\{0^n 1^n \mid n \in \mathbb{N}\}$ is not regular. Let $S = \{0^i \mid i \in \mathbb{N}\}$. Then, for any two elements $x, y = 0^j, 0^k$ in S with $j \neq k$, $z = 1^j$ gives $xz \in L$, but $yz \notin L$.

Note that, if you are trying to prove a language regular, there is almost always a simpler method available than the Myhill-Nerode theorem. Similarly, to prove the non-regularity of a language on exams and homeworks, we encourage you to use the pumping lemma rather than the Myhill-Nerode theorem. It is a powerful tool, but can be difficult to use correctly.

1.4 Problems

Turn in number 1 and two of the remaining problems.

1. Prove that $\sim_L$ is an equivalence relation.

Proof: We need to show reflexive, symmetric and transitivity. Let $\sim_L = R$

- (a) Reflexive: We need xRx . Clearly for any given string $z \in \Sigma^*$ we have $xz \in L \iff xz \in L$ as they are both the same statement. So trivially xRx .
- (b) Symmetric: We need to show $xRy \implies yRx$. If xRy then for any $z \in \Sigma^*$ we have $xz \in L \iff yz \in L$, but this is equivalent to $yz \in L \iff xz \in L$ (as we have if and only if), but this is just yRx by definition. Hence, we showed $xRy \implies yRx$.
- (c) Transitive: We need xRy and yRz implies xRz . Let $k \in \Sigma^*$ be our arbitrary string. Clearly xRy means $xk \in L \implies yk \in L$ similarly yRz means $yk \in L \implies zk \in L$.

⁴You should see this is regular quickly as this is just $(11)^*$, and existence of a regular expression implies it is regular.

Since they are all if and only if, it means they are all equivalent statements (more precise we can show $xk \in L \implies yk \in L \implies zk \in L$ and the reverse as well). Hence we have $xk \in L \iff zk \in L$ which is just xRz . ■

2. Give an example of a non-regular language that cannot be proved to be non-regular using the pumping lemma. (Prove your work, do not just state the language.)
3. Let $L_{n,c} = \{1^k \mid k \equiv c \pmod{n}\}$. Prove this language is regular for any c, n WITHOUT constructing a DFA.

Proof: We have the equivalence relation $\sim_L$ on Σ^* . Now consider the following partition $[0], \dots, [n-1]$ where for a string $x = 1^a \in \Sigma^*$ we have $x \in [k] \iff a \equiv k \pmod{n}$. Now note that we have $\Sigma^* = [0] \cup \dots \cup [n-1]$ as for any strings 1^k we have some $j \in \{0, \dots, n-1\}$ for which $k \equiv j \pmod{n}$ hence $1^k \in [j]$ uniquely. So we need to show two elements in an equivalence class are equivalent. Consider for any $0 \leq j \leq n-1$ the class $[j]$ note that if $x = 1^a, y = 1^b \in [j]$ then we have $a \equiv j \pmod{n}$ and $b \equiv j \pmod{n}$. So note that for any string $z = 1^d$ we have $xz \in L \implies 1^{a+d} \in L \implies a+d \equiv c \pmod{n} \implies b+d \equiv c \pmod{n} \implies yz \in L$ similarly if $xz \notin L$ we have $a+d \not\equiv c \pmod{n} \implies b+d \not\equiv c \pmod{n}$ which means $yz \notin L$. Hence we show that $x \sim_L y$. So we have partitioned our set Σ^* under the relation $\sim_L$ into a finite number of equivalence classes. ■

4. Prove that the DFA from the proof in 1.2 is minimal (i.e. there is no DFA for L with fewer states.)
5. Let $L = \{1^{n^2} \mid n \in \mathbb{N}\}$. Prove that L is not regular using the theorem.

Proof: We need an infinite set S of strings such that for each distinct pair $x, y \in S$ for some z we have $xz \in L$ and $yz \notin L$. Simply consider $S = L$. Now note that for any $x, y \in L$ we have $x = 1^{n^2}, y = 1^{m^2}$ for some $n \neq m \in \mathbb{N}$. Now WLOG assume $n < m$ and take $z = 1^{2n+1}$. Now note that we have, $xz = 1^{n^2}1^{2n+1} = 1^{n^2+2n+1} = 1^{(n+1)^2}$ and $yz = 1^{m^2}1^{2n+1} = 1^{m^2+2n+1}$. Clearly $xz \in L$ and $n \in \mathbb{N} \implies n+1 \in \mathbb{N}$. Now we need to show that $1^{m^2+2n+1} \notin L$. Note the next square after m is $(m+1)^2 = m^2 + 2m + 1$. Now clearly there cannot be any square between m^2 and $(m+1)^2$. But if $m > n$ then $2m+1 > 2n+1$ which implies that $m^2 + 2m + 1 > m^2 + 2n + 1$. This gives us,

$$m^2 < m^2 + 2n + 1 < m^2 + 2m + 1 = (m+1)^2$$

This means there is no natural number whose square gives us $m^2 + 2n + 1$ which implies that $1^{m^2+2n+1} \notin L$. Hence for any distinct x, y we found a z such that neither are in L or outside L at the same time. This concludes our proof. ■

Of additional interest to you may be problems 1.47, 1.48⁵ from the Sipser book.

⁵1.34, 1.35 in the first edition